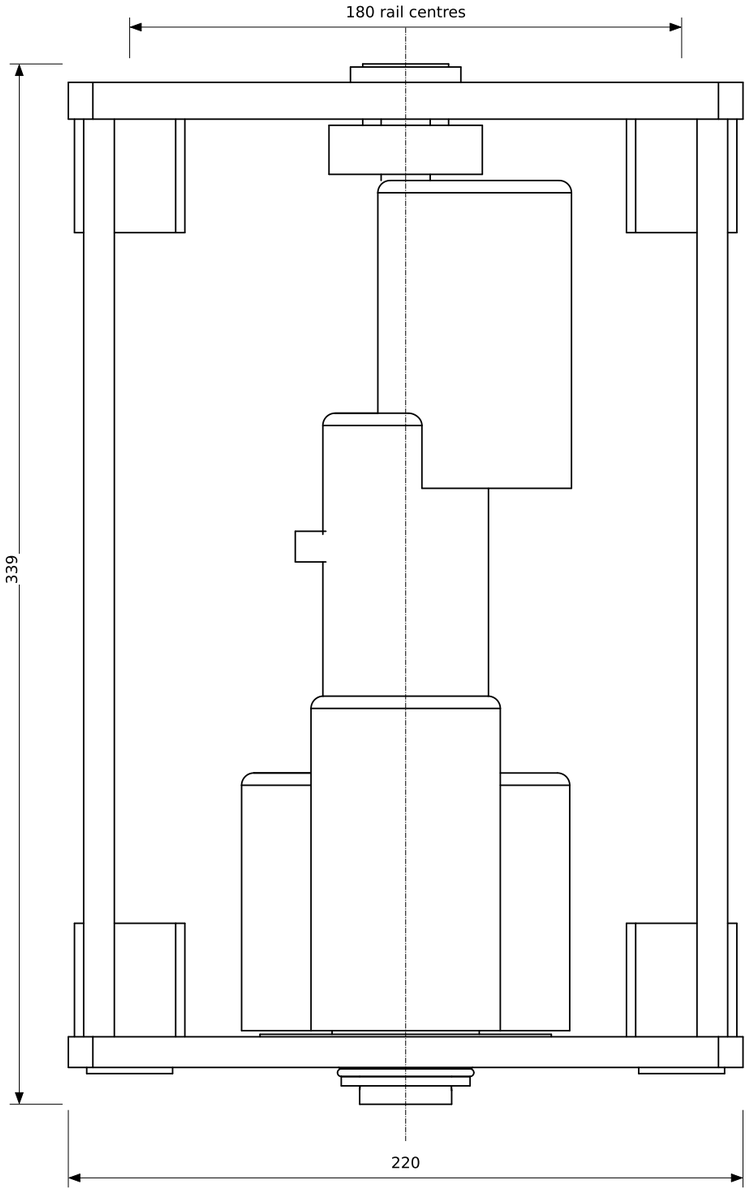
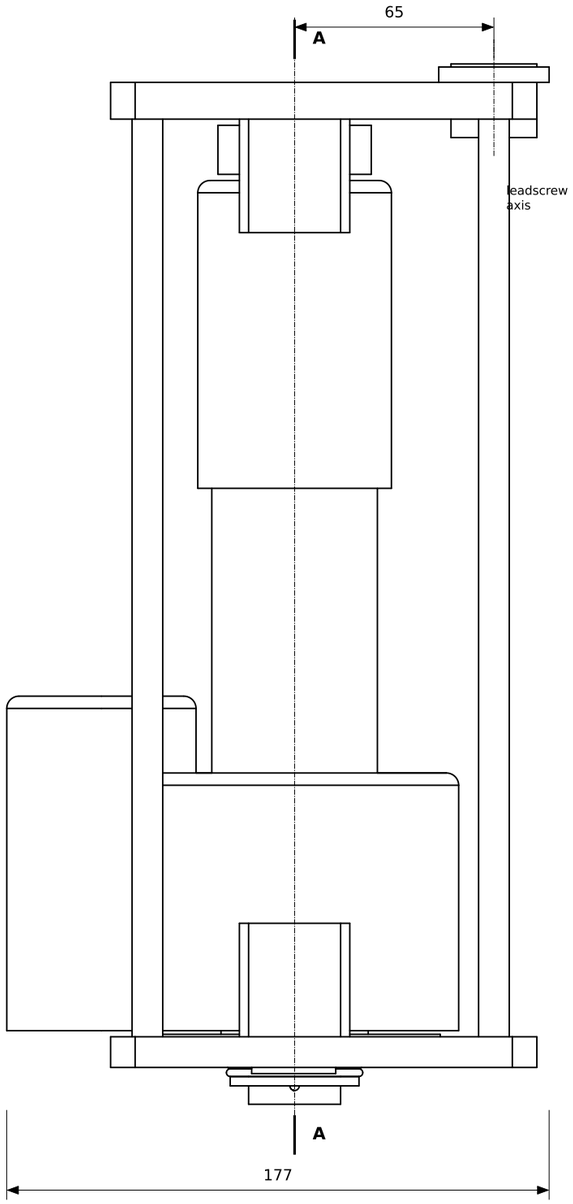
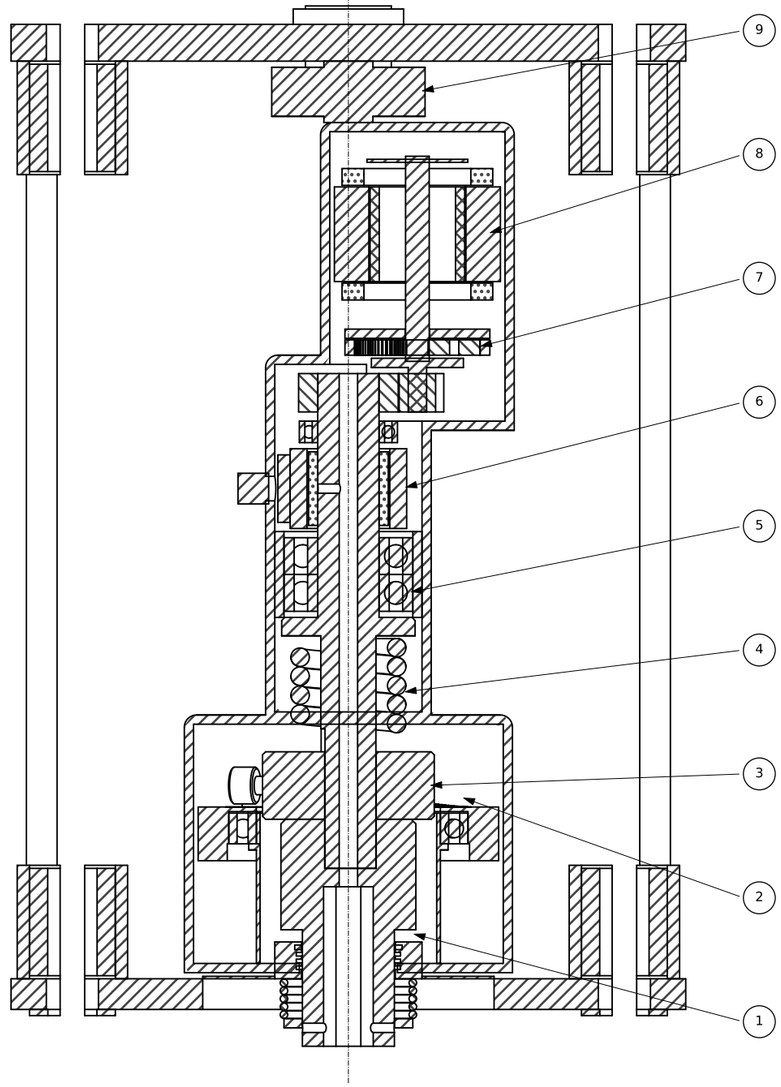
FRONT VIEW



RIGHT SIDE VIEW



SECTION A-A (spindle axis)



NOTES

1. Rotary: frameless BLDC -> 12/36/84 planetary (3 planets) -> 12T/24T spur; spindle on SKF 7004 ACD pair.
2. Percussion: 2-lobe barrel cam lifts the 0.379 kg hammer against a 6 mm 17-7PH spring; 2.25 J at 26 Hz.
3. Feed: Tr16x4 bronze nut, self-locking; WOB measured by the 500 N load cell under the yoke.
4. Housing Ti-6Al-4V; labyrinth + PTFE lip seals and dust bellows exclude regolith.
5. Bearing lives and gear checks: see the design-derivation ledger. General tolerances ISO 2768-mK.

ITEMS (section A-A)

1 Anvil / hex socket	4 Hammer spring	7 Planetary
2 Barrel cam	5 Angular-contact pair	8 Rotary BLDC
3 Hammer	6 Slip ring	9 WOB load cell

ENCLOSED ROTARY-PERCUSSIVE DRILL HEAD

Autonomous low-gravity rotary-percussive bite-drilling system

Material: Housing Ti-6Al-4V; gears 9310	Scale: 1:2 Units: mm Sheet: A3	
Drawn: Jay Tilak P. Date: 2022-09-07	DWG NO. **DRL-002**	REV A